

Person re-identification from virtuality to reality via modality invariant adversarial mechanism

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ABSTRACT

Person re-identification based on multi-style images helps in crime scene investigation, where only a virtual image (sketch or portrait) of the suspect is available for retrieving possible identities. However, due to the modality gap between multi-style images, standard model of person re-identification cannot achieve satisfactory performance when directly applied to match the virtual images with the real photographs. To address this problem, we propose a modality invariant adversarial mechanism (MIAM) to remove the modality gap between multi-style images. Specifically, the MIAM consists of two parts: a space transformation module to transfer the multi-style person images to a modality-invariant space, and an adversarial learning module “played” between the category classifier and modality classifier to steer the representation learning. The modality classifier discriminates between the real and virtual images while the category classifier predicts the identities of the input transformed images. We explore the space transformation for data augmentation to further bridge the modality gap and facilitate the performance. Furthermore, we build two new datasets for the multi-style Re-ID to evaluate the performance. Extensive experimental results demonstrate the effectiveness of the proposed method on improving the performance against the existing feature learning networks. Further comparison results conducted on different modules in MIAM show that our approach is of favorable generalization ability on alleviating the modality gap to improve the multi-style Re-ID.

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1. Introduction

Person re-identification is of important capability for surveillance systems [1–3]. Traditional person Re-ID aims to match images of the same identity across different non-overlapping camera views. It remains a challenging issue due to illumination and pose variations among different views. However, there are many cases where law enforcement agencies have no photos of a suspect and only a virtual image (sketch or portrait) made with the help of an eyewitness or victim is available. Furthermore, with the popularity of image processing software, there remains an urgent demand of Re-ID between the artificial image and the real person image. We describe the problem as multi-style person Re-ID from virtuality to reality, where “virtuality” denotes the artificial images with artistic treatment, and “reality” denotes the natural images captured by real surveillance systems.

The traditional Re-ID suffers severe variations across different camera views. Compared to the traditional person Re-ID task, multi-style Re-ID contains the problems existing on the traditional Re-ID and brings more violent challenges. Due to significant differences with real images from surveillance systems, the virtual images cannot be easily matched with the real identity by the traditional recognition methods [4]. This problem has been defined in the literature as modality gap [5]. As shown in Fig. 1, besides variations existing among different camera views, the modality gap between the different image styles brings extra challenges that limit the performance. As different data sources typically have different statistical properties and distributions, it is difficult to directly compare each other for feature matching [6].

One solution to solve the modality gap between different data sources is doing data augmentation across sets, such as using cycleGAN [7] to do image transformation across different camera views [8] or datasets [9]. However, the fixed data augmentations scheme can not supply flexible input changes to help further promote feature learning. Other representative method includes pre-training the source encoder to adjust the target encoder that can

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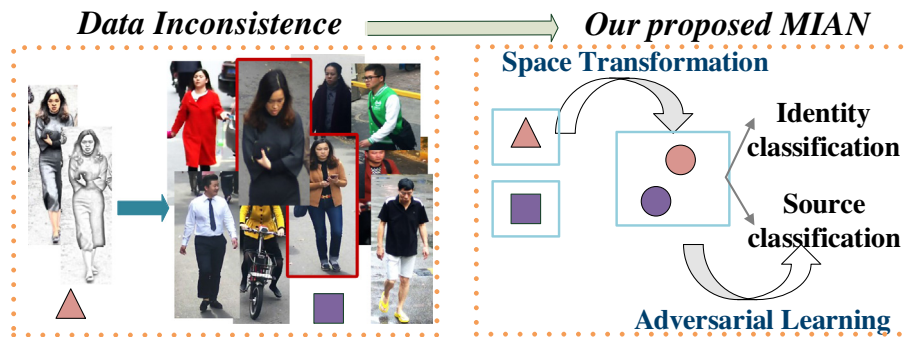


Fig. 1. Challenges for multi-style person re-identification and the structure of the proposed MIAM. Besides different surveillance camera views variations, the modality gap between the different image styles brings more violent challenges. We propose a modality invariant adversarial mechanism incorporating space transformation and adversarial learning to deal with the challenges in multi-style Re-ID, improving the performance of existing feature learning networks.

not be discriminated from each other [10]. The fixed classifier trained on the source domain for the target classification also lack of generalization for the recognition across domains [10]. Some other works [11,6] propose an adversarial learning network on the feature plane to enable flexible retrieval experience across different modalities. They usually need pre-trained feature extractor to obtain good performance, which limit their practicability. Furthermore, we observe that adversarial learning on the feature plane cannot well solve the modality gap, as the high-level features always lack the underlying information of original data. Therefore, in this paper, we explore to bridge the modality gap in the original data plane, and propose an end-to-end framework that incorporates the data transformation with the classification to further solve the multi-style Re-ID task.

To address the above-mentioned problems, we propose a modality invariant adversarial mechanism (MIAM) to handle the multi-style Re-ID task. Two new datasets special for the problem are proposed, the sources of which are from constructed real surveillance systems. The MIAM is to deal with the challenges caused by multi-style data sources, improving the performance of existing feature learning networks. From our observation, person Re-ID on different data sources with multiple image styles, can be regarded as a cross-modal task. We treat the data sources as different modalities to solve these problems by bridging the modality gap in a unified network. Specifically, the proposed MIAM firstly adopts a *space transformation* module to transfer the data from different sources into a modality-invariant space and remove the inconsistency caused by the modality gap in multi-style Re-ID. Then the *adversarial learning* “played” between the category classifier and the modality classifier is introduced to steer the representation learning, where the modality classifier discriminates between the real and virtual images and the category classifier predicts the identities of the input transformed images. We explore the space transformation for data augmentation to further bridge the modality gap. We also conduct the performance on different improved architectures to demonstrate the generalization of our method. The proposed datasets and MIAM model will be made available to the public.

The contributions of our work are summarized as follows: (1) Propose a modality invariant adversarial mechanism (MIAM) incorporating space transformation and adversarial learning to address the modality gap for improving multi-style Re-ID; (2) Propose two new datasets from virtuality to reality for multi-style Re-ID; (3) Explore space transformation for data augmentation and different architectures of the category classifier for further performance improvement.

2. Related work

Existing approaches for the traditional Re-ID can be divided into two aspects: feature [12–20] or distance learning [21–26] to directly learn projections of data items from different camera views into a common feature representation subspace in which the similarity between them can be assessed directly. However, the over-fitting usually occurs when the system is not of good generality to out-of-set examples, especially on the challenging datasets which contain large variations across different camera views. To relieve the over-fitting, some works adopt multiple attention mechanism to make the system focus on more crucial area. The work of Ref. [27] utilizes the quality-aware network model to pay more attention to the images of high-quality. The work of Ref. [28] proposes to select the most discriminative frames of the person as the input for recognition. These methods achieve considerably improved performance. However, these methods only consider the spatial features which easily suffer from the camera view variations. Some works [29,30] utilize the deep recurrent network (RNN or GRU) for the video-based person re-identification to extract the more reliable temporal feature. The work proposed by [31] utilizes attention model to pay emphasis on the more important area and frames to make the learned feature by the RNN more effective. These methods achieve considerable improvements of performance since the more reliable temporal feature is utilized. However, the data inconsistency caused by the view gap also exist to limit the performance of existing networks.

Compared to the traditional Re-ID, the modality gap [5] in the multi-style Re-ID leads to more difficulty for direct comparison with each other by these feature matching methods. Existing networks for the traditional Re-ID can not be well applied on the multi-style Re-ID task, since these methods do not change the distribution of data sources to remove the modality gap. To make the feature extracted by these methods comparable, the modality gap in the data sources should be well solved for improving the multi-style Re-ID. A similar problem with the multi-style Re-ID is the face sketch recognition, as it also works on solving the modality gap problem. A variety of face sketch recognition methods have been proposed recently to learn the discriminative feature representation for solving this problem, including utilizing more discriminative traditional feature [4], useful feature fusion [32] and deep learning based method [33] to bridge the modality gap between sketches and multi-shot photos. However, these methods are targeted for specific image style. Since the multi-style Re-ID typically has different statistical properties, the face sketch recognition methods are of limited generalization in the task.

Some other approaches to solve the modality gap are the cross-modal retrieval methods [34–36], such as from text, speech to image retrieval. Cross-modal retrieval aims at retrieving relevant items that are of different nature with respect to the query format. The main challenge is to measure the similarity between different modalities of data. Existing approaches for the cross-modal retrieval usually map images and other modality data into a shared latent space in which they can be compared. The work of Ref. [37] is based on a deep autoencoder network. Multiple instances are input into the network to learn common representation at the joint layer, which also has the ability to reconstruct both modalities. The work of Ref. [38] consists of two autoencoder networks coupled at the code layer to simultaneously model the reconstruction error and correlation loss. The work of Ref. [39] jointly learns basis matrices of different modalities, by using a local group based priori in the formulation to fully take advantage of popular block based features. The work of Ref. [40] adopts multiple deep networks to generate separate representation and learns common representation with a stacked network. These methods aim at learning a metric in that the feature from different modalities can be comparable. However, they only consider to solve the modality gap in the feature plane, but ignore to pre-remove the gap caused by the data, which limits the performance of these methods on the multi-style Re-ID task. The work of Ref. [6] proposes an adversarial cross-modal retrieval network to enable flexible retrieval experience across different modalities, achieving considerable performance in the cross-modal retrieval tasks. However, they need pre-trained feature extractors, which limits its performance on the multi-style Re-ID problem containing severe modality gap by different data sources. How to solve the multi-style Re-ID problems containing severe modality gap remains challenging.

3. Methodology

3.1. Overview

In this section, we concretely introduce the framework of our MIAM network. As shown in Fig. 2, the MIAM includes space transformation and adversarial learning for simultaneously removing the modality gap and improving the performance of existing feature learning networks. The space transformation module consists of an image generator, trying to transfer the original images from two inconsistent sources into a modality-invariant space. The adversarial learning plays between a category classifier and a modality classifier for adjusting the space transformation and improving the performance of existing Re-ID models. The modality classifier discriminates between the real and virtual images while the category classifier predicts the identities of the input transformed images. To further improve the performance, space

transformation for data augmentation is adopted to further bridge the modality gap.

3.2. Space transformation

Since the severe modality gap exists in multi-style Re-ID, it is difficult to directly compare the features of the original images extracted by traditional Re-ID methods. To make them directly comparable, we aim at learning the transformed space “S” for different modalities to make the distributions of “A” and “B” transformed to be modality-invariant and semantically discriminative. In this section, we propose a data transfer generation method to bridge the modality gap. We utilize a 9-layer resnet generator [7] (denoted as “G”) to operate on the images from two modalities and remove the inconsistency by learning the consistent and discriminative transferred images. Specifically, assuming that there is a collection instance of image pairs from data source “A” and data source “B”, a and b are the images in the corresponding sources, i.e. $a \in A, b \in B$. The desired common subspace S is the transformation from modality A and modality B , which can be projected as $S = f(A, B; \theta_G)$. Here, f is the mapping function to transform the images to a common space S . During training, for a batch size n , every batch input images $I = \{a^i, b^i\}_{i=1}^n$, $a \in A, b \in B$, generate corresponding images $\hat{I} = \{\hat{a}^i, \hat{b}^i\}_{i=1}^n$, $\hat{a}, \hat{b} \in S$ by the network “G”. The parameter denoted as θ is to adjust the mapping procedure, then the operation of the network can be computed as:

$$\hat{I} = G(I; \theta_G), \quad (1)$$

where the G denotes the procedure by the mapping generator, equivalent to the above-mentioned mapping functions f . Then the transformed images are sent to the next step for further recognition. In the following subsections, we will explain how to meet the requirements that the distributions of “A” and “B” are transformed to be modality-invariant and semantically discriminative, while cross-modal similarity structure in data is preserved.

3.3. Modality classifier

We then utilize the modality classifier, referred as “D”, to help improve the generated modality-invariant space “S”. Recently, the “D” widely acts as “discriminator” in existing GAN network [41–43], discriminating the sources of the generated data. In this paper, we introduce the modality classifier to act as an adversary for the space transformation, helping the “G” generate more suitable mapping function that can confuse “D”. In MIAM, the top of the modality classifier is activated by the cross-entropy loss. We manually assign two labels to act as the ground truth of the sources on these images. The images generated from data source A are

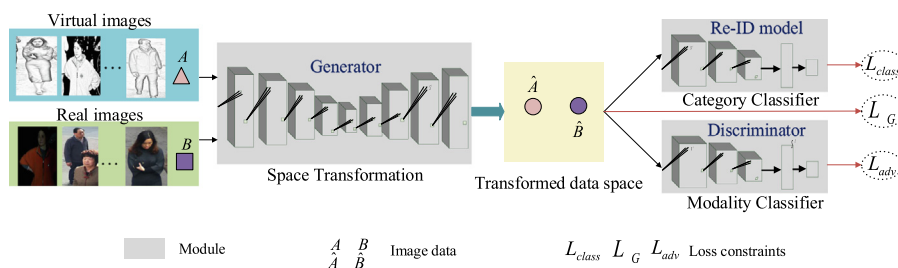


Fig. 2. The diagram of our proposed MIAM model. It includes space transformation and adversarial learning for simultaneously removing the modality gap and improving the performance of existing feature learning networks. We use an encode-decode generator to do the space transformation for transferring the data from different inconsistent sources (“A”, “B”) to a modality-invariant spaces “S”; Adversarial learning plays between a category classifier and a modality classifier for adjusting the generator and improving the performance of existing Re-ID model for obtaining a more discriminative features extractor that is beneficial to multi-style Re-ID.

uniformly labeled as 0 while the images generated from data source B are uniformly labeled as 1. Therefore, the adversarial loss L_{adv} is defined as:

$$L_{adv} = -\frac{1}{n} \sum_{i=1}^n \left(m^i \cdot \log D(\hat{a}^i; \theta_D) + m^i \cdot \log (1 - D(\hat{b}^i; \theta_D)) \right). \quad (2)$$

Essentially, on training, L_{adv} denotes the cross-entropy loss of two categories in modality classification for all instances $\{\hat{a}^i, \hat{b}^i\}_{i=1}^n$ used per iteration. Furthermore, m^i is the ground truth modality label of each instance, while $D(\cdot; \theta_D)$ is the generated modality probability (from A or B) of the instance. The adversarial loss L_{adv} is fed back to the space transformation network, to adjust the generator for improving the mapping function.

3.4. Category classifier

After the images from different modalities transferred into a common modality-invariant space “ S ”, in the second step, the adversarial learning “played” between the category classifier (denoted as “ L ”) and modality classifier (denoted as “ D ”), is introduced to steer the representation learning. As shown in Fig. 2, we model “ L ”, “ D ” and their interaction to effectively and efficiently meet the requirements that generated images from the distributions of A and B should be modality-invariant and semantically discriminative. In this section, we utilize different “ L ” to be recognition networks for measuring the efficiency of the space transformation network. In our case, the “ L ” comes from the latest classification networks with the top activated by multiple loss constraints. To bridge the modality gap, we propose the classification constraint applied to each modality helps to learn more discriminative features.

First we use the softmax to make the label prediction of the transformed images in the common subspace “ S ” be discriminative under given the semantic labels. The classifier takes the batch of transformed instances $\hat{I}$ from different modalities as coupled training data and generates a probability distribution output of semantic categories, the corresponding label of which is $c = \{c_a^i, c_b^i\}_{i=1}^n$. With the softmax loss, we use these probability distributions $\hat{p}$ to formulate the discrimination loss as follows:

$$L_{label} = -\frac{1}{2n} \sum_{i=1}^n \sum_{j=1}^n \left(c_a^i \log \hat{p}_i(\hat{a}^i) + c_b^j \log \hat{p}_j(\hat{b}^j) \right), \quad (3)$$

The L_{label} denotes the computed loss of semantic category classification with all the instances $\hat{I} = \{\hat{a}^i, \hat{b}^i\}_{i=1}^n = G(\{\hat{a}^i, \hat{b}^i\}_{i=1}^n; \theta_G)$, where n is the batch size. Here, in a mini-batch, $\hat{a}^i$ from A and $\hat{b}^i$ from B are randomly selected respectively, thus the ground truth of each instance from different modalities is random. Then the probability distribution $\hat{p}_i, \hat{p}_j$ is generated for each item in A or B at a mini-batch respectively.

The triplet loss is also utilized to pulling the instances of the same person closer and pushing the instances of different persons apart.

$$L_{triplet} = \sum_i \left[\|\hat{a}_a^i - \hat{b}_p^i\|_2^2 - \|\hat{a}_a^i - \hat{b}_n^i\|_2^2 + \alpha \right], \quad (4)$$

where the α of the triplet loss is 1.0. The anchor $\hat{a}_a^i$ is from the A , the positive sample $\hat{b}_p^i$ and the negative sample $\hat{b}_n^i$ of which is from the B . Then the global loss function is calculated as:

$$L_{class} = W_0 L_{label} + W_1 L_{triplet}, \quad (5)$$

where the loss weight W_0, W_1 is 1.0. The loss of the matching probability is back-propagated to adjust the parameters of the category

classifier for mixing the modalities of the $\hat{a}^i$ and $\hat{b}^i$. Moreover, it is also back-propagated to the space transformation network to adjust the generator for improving the mapping function. In this paper, we utilize different “ L ” to be the recognition networks for measuring the efficiency of the space transformation network. Similar to “ D ”, the classification loss is also added to be fed back to the space transformation network, to adjust the generator for improving the mapping function. The full is formally defined as:

$$L_G = \alpha L_{class} + (-\beta L_{adv}), \quad (6)$$

where the hyper-parameters α and β control the contributions of the two terms. In our case, we empirically set the α as 1.0 and β as 2.0.

3.5. Optimization

Algorithm 1. Pseudocode of optimizing MIAM.

Initialization: Randomly selected image pairs for current

batch: $I = \{\hat{a}^i, \hat{b}^i\}_{i=1}^n$; Corresponding labels for current

batch: $l = \{l_a^i, l_b^i\}_{i=1}^n$; hyperparameters: α, β ; n samples in mini-batch for each modality A and B ;

update until convergence:

1: update parameters θ_G, θ_{label} by descending their stochastic gradients:

2: $\theta_G \leftarrow \theta_G - \mu \cdot \nabla_{\theta_G} \frac{1}{2n} (\alpha L_{class} - \beta L_{adv})$;

3: $\theta_{label} \leftarrow \theta_{label} - \mu \cdot \nabla_{\theta_{label}} \frac{1}{2n} (L_{class})$;

4: update parameters of modality classifier by descending their stochastic gradients:

5: $\theta_D \leftarrow \theta_D - \mu \cdot \nabla_{\theta_D} \frac{1}{2n} (L_{adv})$;

6: **return** learned representations in common subspace:

$f(G(A, B; \theta))$

The process to learn the optimal feature representation of MIAM is conducted by jointly minimizing the adversarial and category losses obtained above. Since the optimization goals of these two objective functions are opposite, the process runs as a minimax game of the two concurrent sub-processes:

$$(\theta_G, \theta_{label}) = \arg \min_{\theta_G, \theta_{label}} (\alpha L_{class} - \beta L_{adv}), \quad (7)$$

$$\theta_D = \arg \min_{\theta_D} L_{adv}. \quad (8)$$

This minimax game is implemented using the optimization algorithm “Adam”. The minimax optimization is alternately performed, as shown in the Algorithm 1.

3.6. Data augmentation

To further improve the performance of the MIAM model, we adopt the space transformation for data augmentation.

As shown in Fig. 3, we adopt a space transformation method for data augmentation to further relieve some influence caused by the large view variations in vision. Given one set of images in camera A and a different set in camera B , the goal of the data transformation is to train a mapping $E: A \rightarrow B$ such that the output $\hat{b} = E(a), a \in A$, is similar to images $b \in B$ to remove the variation. Since the Re-ID problem is usually not of one-to-one correspondence, the work by [7,8] inspired us to adopt “cycle consistent” structure to reduce one-to-one constraints and perform unpaired data transformation especially for the Re-ID task. Specifically, setting two mappings

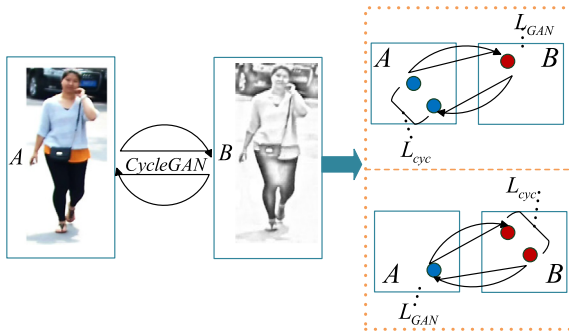


Fig. 3. Data transformation for data augmentation. We utilize the cycleGAN [7] to transfer the real image to the virtual image and add the transferred generated images to the original data for augmentation.

that are bijections denoted as E and F , translator $E: A \rightarrow B$ is inverses of another translator $F: B \rightarrow A$. This structural assumption is applied by training both the mapping E and F simultaneously, and adding a cycle consistency loss that encourages $F(E(a)) \approx a$ and $E(F(b)) \approx b$. Combining this loss with adversarial losses on domains A and B yields the full objective for unpaired space transformation.

Similar to the work of Ref. [8], we utilize the mapping “E” to transfer the real image to virtual image and add the transferred generated images to the original data for augmentation, pre-removing the style variation in the data plane. However, the data transformation is unsupervised without semantic label constraint. Our MIAM model then is processed on the augmented data to further bridge the modality gap for improving recognition. Table 1.

4. Experiments and results

To clarify the effectiveness of our method, we propose two new datasets for the virtuality to reality Re-ID problem. In this section, detailed experimental results conducted on the two datasets are given to show the performance of our method. We explore different architectures of the category classifier in MIAM to give comparisons of the performance on improving existing feature learning networks. Further comparison results conducted on different modules in MIAM are also given to show the generalization ability of our method on bridging the modality gap in multi-style Re-ID.

4.1. Datasets

In this paper, we propose two new datasets special for the virtuality to reality Re-ID problem, “Sketch2Reality” and “Portrait2Reality”, to elaborate challenges and evaluate the performance of our method. The real image source of these datasets is from constructed real surveillance systems in residential areas. The virtual images are processed by the image processing software “Mtxixiu”. The examples of our proposed datasets are shown in Fig. 4.

Sketch2Reality: The Sketch2Reality dataset is created for matching between single sketch image and multiple real person images. Real image data contains 245 people with totally 40,971 images detected from multiple different camera views (from front to side) and times (from day to night). The number of the camera view is from 3 to 8. Lighting variations, appearance change as well as severe illumination, exist in the data. Every person contains the variable number of images, from 35 to 560. Each person corresponds to a fixed sketch which is randomly selected from the real person images and obtained by the image processing software. The selected real images which are used to generate the virtual images are removed from the corresponding set. Examples are shown in Fig. 4. The dataset is challenging due to insufficiency caused by the poor number of images and severe modality gap caused by a single sketch to varied actual person Re-ID.

Portrait2Reality: Similar to the Sketch2Reality dataset, we propose another dataset from single person portrait to multiple real person images retrieval, real image data of which is the same as the Sketch2Reality dataset. As shown in Fig. 4, each person corresponds to a fixed portrait which is randomly selected from the real person images and obtained by the image processing software. Compared to the Sketch2Reality dataset, the dataset is less challenging since it contains more information for Re-ID.

4.2. Experimental setup

We randomly select approximately half of the persons for training and the others for testing. Specifically, there are 123 identities for training and the other 122 identities for testing on our proposed virtuality2Reality datasets. For training, images are resized to 143×143 and randomly cropped to 128×128 for sending to MIAM, while for testing, images are resized to 128×128 for end-to-end space transformation and recognition. Each training procedure takes more than 10 epochs, and more than 10 times to obtain an average and representative results. In MIAM, the “G” and “D” refer to the generator and discriminator network proposed by Ref. [7]. Different architectures of “L” is explored to show the performance of MIAM in improving the effectiveness of existing feature learning networks, including the latest recognition networks MobileNetV2 [44], ShuffleNet [45], and ResNet50 [46]. We also adopt the state-of-the-art Re-ID network MCDIM [47] to act as the baseline of “L”, which also addresses the cross-domain instance matching problem.

Furthermore, we conduct the experiments with our proposed two datasets on different kinds of state-of-the-art methods, including the latest recognition networks [44–46], cross-domain instance matching [47], harmonious attention network for Re-ID [48], and latest multi-scale deep architecture for Re-ID [49] to comprehensively compare with our MIAM networks. The matching rate is based on the Euclidean Distance. We adopt widely utilized Cumulative Match Characteristic (CMC) curve along with the Mean Average Precision (mAP) to fully show the performance of our proposed method.

Table 1

The characteristics of our proposed datasets for multi-style person re-identification, where the “True” denotes the existence.

Datasets	Identities	Cameras	Images	Label
Sketch2Reality & Portrait2Reality	245	3–8	40971	Hand
	Crop Size	Multi-shot	Tracking Sequences	Full Frames Availability
	Vary	True	True	True
	Changed Clothes	Pose Variation	Clutters	Lighting Variations
	True	True	True	True
	Different Travel Modes		Parts Variations	
	True		True	

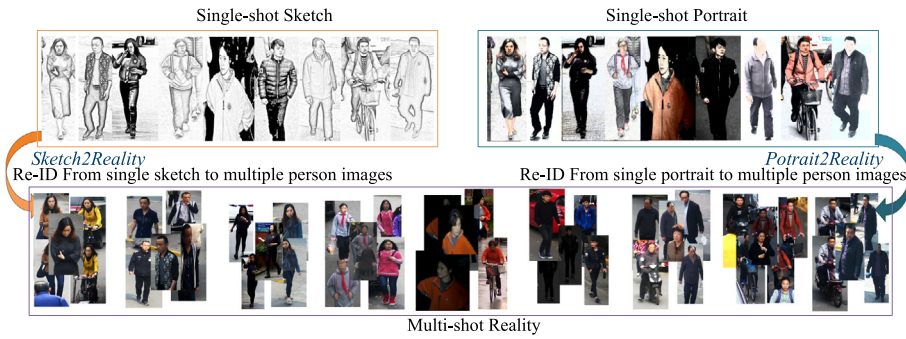


Fig. 4. The examples of our propose two new datasets special for the virtuality to reality Re-ID problem, which are “Sketch2Reality” and “Portrait2Reality”. Compared to the Sketch2Reality dataset, the Portrait2Reality dataset is less challenging since it contains more information for helping the Re-ID procedure.

4.3. Evaluation and results

4.3.1. Multi-style person Re-ID

To show the performance of our MIAM model on removing modality gap among multiple image styles, we set the data from the two different sources as two modalities “A” and “B”, sending them to MIAM with corresponding semantic labels. The virtual data (sketch or portrait) is the “A”, while the real data collected from the real surveillance system is the “B” regardless of the different camera views. After training, we utilize “G” incorporating “L” as a feature extractor deployed on the testing data to compute the CMC and mAP score. The comparison with other state-of-the-art methods is shown on Table 2 and the CMC curves are shown in Fig. 5.

The comparisons are the performance of the state-of-the-art networks deployed on the two datasets to show the performance of MIAM, including latest best classification networks [44–46], cross-domain instance matching [47], harmonious attention network for Re-ID [48], and latest deep multi-scale architecture for Re-ID [49]. Moreover, we also compare the performance before and after data augmentation, where “+ DA” in the Table 2 denotes evaluating the performance after our proposed data balance and data transformation.

In the comparisons, the “L” of our MIAM network on the “Sketch2Reality” is the ResNet50 [46] while that on the “Portrait2Reality” is the MCDIM [47]. From the result, the performance on the Portrait2Reality dataset is better than that on the Sketch2Reality. As shown in Fig. 4, compared to the Sketch2Reality dataset, the Portrait2Reality is less challenging since it preserves more information from the data in the real surveillance system that helps the multi-style Re-ID. The modality gap of the

Sketch2Reality dataset is more severe than the Portrait2Reality, making the performance improvement of the Sketch2Reality more difficult to achieve. As the results, even with state-of-the-art Re-ID network [48,49], the performance on the multi-style Re-ID is limited. Compared to other networks which do not address the modality gap problem, the better performance of the original cross-domain network [47] demonstrates the urgent demand of solving the modality gap that widely exists in the multi-style Re-ID task.

After the modality invariant mechanism, our MIAM outperforms the cross-domain network [47] on both datasets, showing that the proposed space transformation and adversarial learning procedure can better solve the modality gap problem to help learn more discriminative feature representations. Compared to other networks with attention [48] or multiple-scale learning [49] special for the Re-ID problem, our MIAM also performs favorably against the networks on both datasets, demonstrating the effectiveness of our proposed modality invariant adversarial mechanism. Furthermore, the performance improvements are more obvious in the Sketch2Reality dataset than that in the Portrait2Reality. Since the modality gap in Sketch2Reality is more severe, the results show the crucial role of our approach in alleviating the performance loss caused by the modality gap. Moreover, after adding proposed data augmentation methods, our methods obtain performance improvements on both datasets, showing that the proposed data augmentation is helpful for augmenting useful data on improving the multi-style person Re-ID.

4.3.2. Ablation studies

In this section, we further compare the results of the different procedures in our proposals. First, we compare the performance of the modules in MIAM to elaborate their respective advantages

Table 2

Comparison results between our MIAM network and other state-of-the-art approaches, where the top 1 and 2 results are in red and blue (%).

datasets	Sketch2Reality					Portrait2Reality				
	CMC				mAP	CMC				mAP
	R=1	R=5	R=10	R=20		R=1	R=5	R=10	R=20	
MobileNetV2 [44]	15.7	33.7	40.4	57.3	24.2	37.1	60.7	69.7	84.3	47.3
ShuffleNet [45]	28.1	42.7	51.7	71.9	35.8	49.4	71.9	78.7	84.3	60.7
HACNN [48]	28.1	46.1	57.3	66.3	36.7	66.3	80.9	86.5	91.0	73.9
MuDeep [49]	33.7	60.7	73.0	76.4	47.1	70.8	84.3	88.8	91.0	77.0
ResNet50 [46]	41.6	57.3	67.4	84.3	49.9	68.5	86.5	93.3	93.3	76.7
MCDIM [47]	42.7	64.0	75.3	84.3	52.1	71.9	84.3	91.0	93.3	78.0
MIAM	48.3	66.3	75.3	87.6	57.0	74.2	92.1	98.9	98.9	81.5
MIAM+DA	52.8	74.2	84.3	94.4	61.2	75.3	92.1	94.4	97.8	82.7

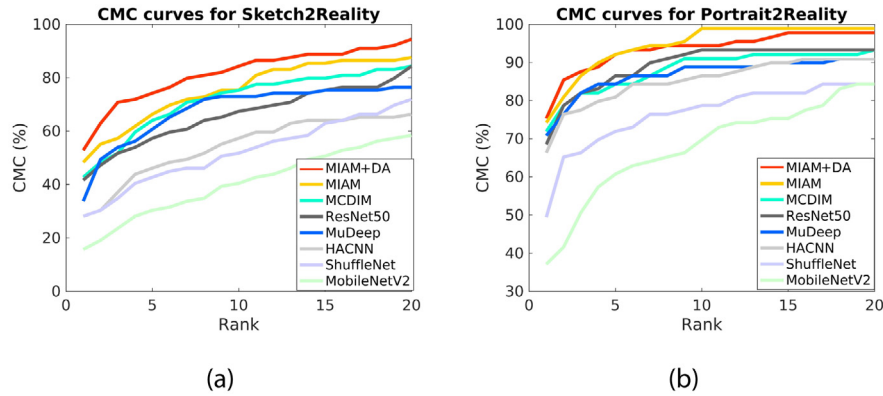


Fig. 5. CMC curves of our proposed MIAM comparing with other state-of-the-art recognition networks deployed on the proposed two virtuality2reality datasets.

and applicability, i.e. “L”, “G”, “D”, showing the effectiveness of the three modules, category classifier, space transformation and modality classifier in our network. Then we explore the different architectures of the “L” in MIAM to show the generalization of MIAM in improving the effectiveness of existing feature learning networks. The baselines include the latest recognition networks MobileNetV2 [44], ShuffleNet [45], ResNet50 [46], and the cross-domain Re-ID network MCDIM [47].

Different Modules in MIAM:

We compare the results of the modules in MIAM to elaborate their respective advantages and applicability. Specifically, we compare the performance of our work that uses the space transformation and adversarial learning for image transformation denoted as “G+D+L”, with existing works that plays the adversarial learning on the feature plane [6] denoted as “D+L”, and the works for traditional Re-ID only with the category classifier denoted as “L”, to show the effectiveness of MIAM. The comparisons are shown in Table 3. We also compare the performance of directly using the cycleGAN [7] to do space transformation on original Re-ID model, analyzing the effect of the data augmentation on feature learning, shown as “L+D”.

From the results, previous cross-modal methods [6] widely utilized the “D+L” network architecture with adversarial learning on the feature plane to solve the modality gap, the effectiveness of which is demonstrated since “D+L” is better than “L”. The adversarial learning can help bridge the modality gap. After adding the “G”, the performance improvement of the “G+D+L” is better than that of the “D+L”, showing that our proposed space transformation procedure is crucially helpful for recognition. The results also demonstrate the effectiveness of our work on transferring the data from different sources into a modality invariant space than directly using the adversarial learning on the feature plane [6].

Results also show that adding the camera transfer data augmentation can help improve the performance of both the single Re-ID model and our MIAM method. We note that “L+DA” performs better than “D+L”, which also demonstrates that bridging

the modality gap on the data plane is better than on the feature plane. Our method “G+L+D” performs better than “L+DA”, shows that our method can supply more flexible and useful transformed data than the fixed data augmentation method to help the multi-style Re-ID task.

Besides, on the “Sketch2Reality” dataset, the performance improvement after adding the “G” is more obvious than that on the “Portrait2Reality”, showing that the transformation procedure has more influence than the adversarial training procedure in the dataset with more severe modality gap, which also demonstrates that the space transformation module “G” plays a crucial and essential role in MIAM for better solving the modality gap. The performance of interplay between the three modules is better than any others, demonstrating the effectiveness of our proposed MIAM model. The results show that MIAM is potentially helpful for solving similar recognition tasks among data sources with different modalities, which will be explored to solve the variations among different camera views on existing Re-ID datasets in the future.

Different Network Architecture Exploration:

To show the generalization of MIAM, we further conduct the experiments with different category classifiers in MIAM on our proposed two datasets, the results of which are shown in Table 4 and Fig. 6. The exploration is made in the “L” since recognition is our goal here. “MIAM1” refers to “L” is MobileNetV2 [44], “MIAM2” refers to “L” is ShuffleNet [45], “MIAM3” refers to “L” is ResNet50 [46], and “MIAM4” refers to “L” is MCDIM [47]. All our MIAM networks here contain no extra proposed data augmentation methods for fair comparisons.

From the result, MIAM all performs favorably against the original classification networks on both datasets, demonstrating that our proposed space transformation and adversarial learning procedure indeed can help the feature learning to improve the performance. Different classifiers show different performances on the two proposed virtuality2reality datasets, where to some content better performance the original network obtain, the better performance is achieved after our interacting the proposed modality

Table 3

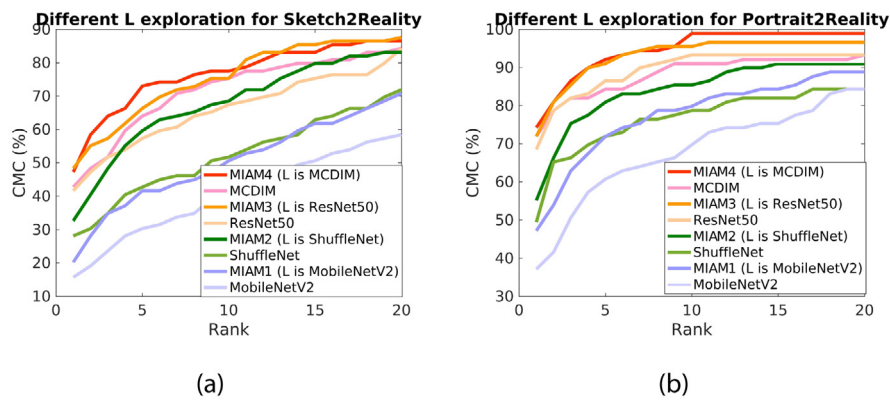
Performance comparison of different modules in our proposed best MIAM method, where for the “Sketch2Reality” that is “MIAM3 (L is ResNet50)” and for the “Portrait2Reality” that is “MIAM4 (L is MCDIM)” (%).

Evaluation	Sketch2Reality					Portrait2Reality				
	CMC				mAP	CMC				mAP
	R = 1	R = 5	R = 10	R = 20		R = 1	R = 5	R = 10	R = 20	
L	41.6	57.3	67.4	84.3	49.9	71.9	84.3	91.0	93.3	78.0
L+DA	45.8	63.5	77.6	87.8	55.7	73.7	92.3	97.0	98.9	81.9
D+L	42.7	59.6	74.2	87.6	52.6	73.0	91.0	95.5	97.8	81.0
G+D+L	48.3	66.3	75.3	87.6	57.0	74.2	92.1	98.9	98.9	81.5
G+D+L+DA	52.8	74.2	84.3	94.4	61.2	75.3	92.1	94.4	97.8	82.7

Table 4

Results of our different architecture exploration on the “L” in MIAM for multi-style person re-identification (%).

dataset	Sketch2Reality					Portrait2Reality				
	CMC				mAP	CMC				mAP
	R=1	R=5	R=10	R=20		R=1	R=5	R=10	R=20	
MobileNetV2	15.7	33.7	40.4	58.4	24.2	37.1	60.7	69.7	84.3	47.3
MIAM1 (L is MobileNetV2)	20.2 ^{↑4.5}	41.6	50.6	70.8	31.2 ^{↑7.0}	47.2 ^{↑10.1}	71.9	79.8	88.8	57.6 ^{↑10.3}
ShuffleNet	28.1	42.7	51.7	71.9	35.8	49.4	71.9	78.7	84.3	60.7
MIAM2 (L is ShuffleNet)	32.6 ^{↑4.5}	59.6	68.5	83.1	44.5 ^{↑8.7}	55.1 ^{↑5.7}	80.9	85.4	91.0	66.3 ^{↑5.6}
ResNet50	41.6	57.3	67.4	84.3	49.9	68.5	86.5	93.3	93.3	76.7
MIAM3 (L is ResNet50)	48.3 ^{↑6.7}	66.3	75.3	87.6	57.0 ^{↑7.1}	71.9 ^{↑3.4}	91	95.5	96.6	80.1 ^{↑3.4}
MCDIM	42.7	64	75.3	84.3	52.1	71.9	84.3	91.0	93.3	78.0
MIAM4 (L is MCDIM)	47.2 ^{↑4.5}	73	77.5	86.5	58.2 ^{↑6.1}	74.2 ^{↑2.3}	92.1	98.9	98.9	81.5 ^{↑3.5}

**Fig. 6.** CMC curves of our different architecture exploration of the “L” in MIAM networks deployed on the proposed two virtuality2reality datasets.**Table 5**Evaluation of loss weights (α, β) in Eq. (6). The Re-ID model for the “Sketch2Reality” that is “MIAM3 (L is ResNet50)” and for the “Portrait2Reality” is “MIAM4 (L is MCDIM)” (%).

Datasets (α, β)	Sketch2Reality					Portrait2Reality				
	R = 1	R = 5	R = 10	R = 20	mAP	R = 1	R = 5	R = 10	R = 20	mAP
(1, 1)	40.4	73.0	83.1	91.0	55.6	72.1	92.1	98.9	98.9	80.8
(1, 2)	48.3	66.3	75.3	87.6	57.0	74.2	92.1	98.9	98.9	81.5
(1, 3)	46.1	64.0	74.2	88.8	56.1	71.3	89.9	94.4	95.5	79.0
(2, 1)	41.6	64.0	77.5	91.0	52.2	74.2	93.3	98.9	100.0	81.4
(3, 1)	44.9	75.3	83.1	88.8	57.7	72.0	91.1	97.8	98.9	80.0
(3, 2)	37.1	65.2	77.5	87.6	49.3	73.9	92.1	98.9	98.9	81.2

invariant adversarial mechanism. The difference exists between the performance of the “MIAM3 (L is ResNet50)” and “MIAM4 (L is MCDIM)” on the “Sketch2Reality” dataset, the CMC rank-1 score of the “MIAM3” is better than the “MIAM4” but the mAP is inverted. From the CMC curves shown in Fig. 6, the “MIAM4” contains better convergence even the CMC rank-1 seems a little bit weaker.

Sensitivity of Hyperparameters:

Table 5 further presents the CMC and mean average precision with respect to different loss weights, i.e., α, β in Eq. (6) to evaluate the impact. The results show that our results are stable within a range for both loss weights α, β (e.g. $1 \leq \alpha \leq 3$ and $1 \leq \beta \leq 3$), although both of them should not be too large to overemphasize

the classification and adversarial constraints. Furthermore, results show that the best performance almost is nearly achieved with $\alpha = 1, \beta = 2$. That is, L_{adv} should be paid more attention to teach the MIAM model to learn to generate better domain invariant data for helping bridge the modality gap.

5. Conclusion

In this paper, we focus on solving the modality gap to improve the performance on multi-style person re-identification task. MIAM is proposed to deal with the severe modality gap between multi-style images. The MIAM includes two procedures: the space

transformation (“G”) to transfer the original images from different data sources to a modality-invariant space; the adversarial learning to play between a category classifier (“L”) and a modality classifier (“D”) for adjusting the space transformation and improving the feature learning networks. For further performance improvements, we explore different architectures of the “L” and adopt two data augmentation methods to make our network more effective and robust. Two new datasets are proposed for the multi-style Re-ID to evaluate the performance. With comprehensive experiments conducted on the proposed datasets, our proposed methods perform favorably against other state-of-the-art networks, demonstrating its effectiveness in improving the performance of existing feature learning networks. We give a further comparison to elaborate the respective advantages and applicability of the different modules in MIAM. With favorable generalization, our method is potentially helpful for solving similar recognition tasks with multiple modalities in the future.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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